

Typical Flight Conditions

- Mach: 0.15 - 0.28
- Altitude
 - En route: 20,000 ft
 - Water Pickup: 0 - 8000 ft
 - Dropping: 100 - 115 ft (Above ground level)
- Angle of Attack (AoA)
 - Cruise: $2^\circ - 5^\circ$
 - Approach / landing: $10^\circ - 12^\circ$
 - Stall speed: $15^\circ - 17^\circ$

Chosen Flight Conditions

1. Pre-drop Approach

- Low altitude, low speed, high Lift configuration
- Full tank

⇒ Stability before release

- Alt = 150 ft, 250 ft
- Mach = 0.15, 0.17

$$\text{Weight} = 43,709.14 \text{ lbs}$$

2. Water - drop

- Low altitude, low airspeed
- Half Tank

⇒ Captures mass and CG changing quickly

- Alt = 100 ft, 150 ft
- Mach = 0.14, 0.16

$$\text{Weight} = 35184.57 \text{ lbs}$$

3. Post-drop & Climb out

- Mid Altitude, increasing speed
- Empty tank

$$\text{Weight} = 28660.0 \text{ lbs}$$

⇒ Shows whether aircraft remains stable when empty

- Alt = 300 ft & 500 ft
- Mach = 0.17, 0.25

Flight Condition #1:

Known:
Parameters

$$\left\{ \begin{array}{l} \bullet \text{Weight} = 43709.14 \text{ lbs} \\ \bullet x_{cg} = 25.67 \\ \bullet z_{cg} = 1.2 \end{array} \right.$$

Flight
Conditions

$$\left\{ \begin{array}{l} \bullet \text{Mach} = 0.17 \\ \bullet \alpha = 7^\circ \\ \bullet \text{Alt} = 250 \text{ ft} \end{array} \right.$$

Wing
Geometry
NACA 4417
↓
 $a_0 \approx 2\pi$

$$\left\{ \begin{array}{l} \bullet S_w = 1023.747 \text{ ft}^2 \\ \bullet b_w = 93.84 \text{ ft} \\ \bullet AR_w = 8.6 \\ \bullet \bar{c}_w = 11.61 \text{ ft} \\ \bullet \text{Incidence} = 2.0^\circ \\ AR = \frac{b^2}{S} ; S = \frac{b^2}{AR} \end{array} \right.$$

Horizontal Tail
Geometry

NACA 0012

$$\left\{ \begin{array}{l} \bullet b = 36 \text{ ft} \\ \bullet \bar{c}_T = 8.67 \text{ ft} \\ \bullet AR = 4.23 \\ \bullet S = \frac{36^2}{4.23} = 306.38 \text{ ft}^2 \\ \bullet \text{Incidence} = 1.0^\circ \end{array} \right.$$

$C_{L\alpha}$

$$C_{L\alpha} = \frac{2\pi AR}{2 + \sqrt{\frac{AR^2 \beta^2}{k^2} \left(1 + \frac{\tan^2 \Lambda_{c/2}}{\beta^2} \right) + 4}}$$

$$\beta = \sqrt{1 - M^2} = \sqrt{1 - 0.17^2} = 0.9854$$

$$k = \frac{a_0}{2\pi} \approx \frac{2\pi}{2\pi} = 1$$

$$C_{L\alpha, w} = \frac{2\pi(8.6)}{2 + \sqrt{\frac{(8.6)^2 (0.985)^2}{1} \left(1 + \frac{\tan^2(\phi)}{\beta^2} \right) + 4}} = 5.0466 / \text{rad}$$

$$C_{L\alpha} = 5.0466 \times \frac{\pi}{180} = 0.0881 / \text{deg}$$

C_L

$$L = \frac{\rho}{2} S C_L \rightarrow \text{Since } L = W$$

$$C_L = \frac{W}{\frac{\rho}{2} S} = \frac{W}{\frac{1}{2} \rho V^2 S_w}$$

$$\textcircled{a} \text{ Alt} = 250 \text{ ft} \rightarrow \rho_{\text{air}} = 0.00236 \text{ slug/ft}^3$$

$$V = M_a =$$

$$a = \sqrt{\gamma R T} = \sqrt{(1.4)(287)(287.6)} = 340 \text{ m/s} = 1115.49 \text{ ft/s}$$

$$V = 0.17(1115.49) = 189.6 \text{ ft/s}$$

Table IV
Geopotential Altitude, English Altitudes

Altitude		Temperature		Pressure		Density	
H (ft)	Z (ft)	T (K)	t (°C)	P (mb)	P/P ₀	ρ (kg/m ³)	ρ/ρ ₀
-1000	-1000	290.131	16.981	1.0504 + 3	1.0366 + 0	1.2612 + 0	1.0296 + 0
-900	-900	289.933	16.783	1.0466	1.0329	1.2576	1.0266
-800	-800	289.735	16.585	1.0428	1.0292	1.2539	1.0236
-700	-700	289.537	16.387	1.0391	1.0255	1.2503	1.0206
-600	-600	289.339	16.189	1.0354	1.0218	1.2467	1.0177
-500	-500	289.141	15.991	1.0316	1.0182	1.2430	1.0147
-400	-400	288.943	15.793	1.0279	1.0145	1.2394	1.0118
-300	-300	288.745	15.595	1.0242	1.0108	1.2358	1.0088
-200	-200	288.546	15.396	1.0205	1.0072	1.2322	1.0059
-100	-100	288.348	15.198	1.0169	1.0036	1.2286	1.0029
0	0	288.150	15.000	1.01325 + 3	1.00000 + 0	1.2250 + 0	1.0000 + 0
100	100	287.952	14.802	1.0095	9.9639 - 1	1.2214	9.9708 - 1
200	200	287.754	14.604	1.0059	9.9279	1.2178	9.9416
300	300	287.556	14.406	1.0023	9.8920	1.2143	9.9125
400	400	287.358	14.208	9.9868 + 2	9.8562	1.2107	9.8835
500	500	287.160	14.010	9.9507	9.8206	1.2072	9.8545
600	600	286.961	13.811	9.9147	9.7850	1.2036	9.8256
700	700	286.763	13.613	9.8788	9.7496	1.2001	9.7968
800	800	286.565	13.415	9.8429	9.7142	1.1966	9.7680
900	900	286.367	13.217	9.8072	9.6790	1.1931	9.7393

→ Interpolating for 250 ft

$$C_L = \frac{Z W}{\rho V^2 S_w} = \frac{Z(43709.14)}{(0.00235)(189.5)^2 (1023.947)} = 1.00632$$

$$C_{D\alpha}$$

$$C_D = C_{D0} + \frac{C_L^2}{\pi e A R}$$

$$C_{D\alpha} = \frac{\partial C_D}{\partial \alpha} = \frac{1}{\pi e A R} \frac{\partial (C_L^2)}{\partial \alpha} = \frac{1}{\pi e A R} 2 C_L C_{L\alpha}$$

$e \approx 0.8$ for rectangular wing with $AR = 8.6$

$$C_{D\alpha} = \frac{2 C_L C_{L\alpha}}{\pi e A R} = \frac{2(1.00632)(50465)}{\pi(0.8)(8.6)} = 0.459922...$$

$$C_{D\alpha} = 0.4599 / \text{rad}$$

$$C_{D\alpha} = 0.008202 / \text{deg}$$

$$C_D$$

→ Using $C_{D0} \approx 0.010$

$$C_D = C_{D0} + \frac{C_L^2}{\pi e A R}$$

$$C_D = 0.010 + \frac{1.00632^2}{\pi(0.8)(8.6)} = 0.05685...$$

$$C_D = 0.0569$$

V_{stall}

→ Approximating $C_{L_{\text{max}}} \approx 1.8$

$$L = \bar{q} S C_L \rightarrow L = W$$

$$W = \frac{1}{2} \rho V_{\text{stall}}^2 S C_{L_{\text{max}}}$$

$$V_{\text{stall}} = \sqrt{\frac{2W}{\rho S C_{L_{\text{max}}}}} = \sqrt{\frac{2(43709.14)}{(0.00235)(1023.9)(1.8)}} = 141.7653 \text{ ft/s}$$

$$V_{\text{stall}} = 141.8 \text{ ft/s}$$

$$= 43.22 \text{ m/s}$$

$$= 84.0 \text{ kt}$$

Lift Force

$$L = \bar{q} S C_L$$

↳ assuming steady level flight $L = W$

$$L = 43709.14 \text{ lbf} = 194428 \text{ N}$$

Drag Force

$$D = \bar{q} S C_D$$

$$= \frac{1}{2} \rho V^2 S C_D$$

$$= \frac{1}{2} (0.00236) (189.6)^2 (1023.9) (0.05685) = 2469.259 \text{ lbf}$$

$$D = 2469 \text{ lbf} = 10,983 \text{ N}$$

Downwash Model

$$\frac{dE}{d\alpha} = 4.44 \left[K_A K_\lambda K_H (\cos \Delta C_{lu})^{1/2} \right]^{1.19}$$

$$\bullet K_A = \frac{1}{AR} - \frac{1}{1+AR^{1.7}} = \frac{1}{8.6} - \frac{1}{1+(8.6)^{1.7}} = 0.091143$$

$\lambda = 1$ since $C_r = C_t$

$$\bullet K_\lambda = \frac{10-3\lambda}{7} = \frac{10-3}{7} = 1$$

$$\bullet K_H = 1 - \frac{h_H}{b} \sqrt[3]{\frac{2l_H}{b}}$$

$$l_H = \text{a.c. tail w.r.t a.c. wing} = x_H - x_W = 57 - 27 = 30 \text{ ft}$$

$$h_H = \text{height of H.T w.r.t wings} = z_H - z_W = 11 - 5 = 6 \text{ ft}$$

$$K_H = \frac{1 - \frac{6}{93.84}}{\sqrt[3]{\frac{2(30)}{93.84}}} = 1.08655$$

$$\frac{dE}{d\alpha} = 4.44 \left[(0.09114)(1)(1.08655) \sqrt{\cos(0)} \right]^{1.19} = 0.28337 \dots$$

$$\frac{dE}{d\alpha} = 0.283$$

$$\odot M_{ach} = 0$$

Longitudinal Stability

$$M_{TOT} = M_p + M_w + M_{H.T.}$$

$$(C_{m_\alpha})_{TOT} = C_{m_{\alpha,f}} + C_{L_{\alpha,w}} \bar{x}_a - C_{L_{\alpha,t}} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \eta \bar{v}$$

• Wing Contribution:

$$C_{L_{\alpha,w}} \approx C_{L_\alpha} \rightarrow \text{calculated before}$$

$$C_{L_{\alpha,w}} = \frac{2\pi(8.6)}{2 + \sqrt{\frac{(8.6)^2(0.985)^2}{1} \left(1 + \frac{\tan^2(\phi)}{\beta^2}\right) + 4}} = 5.0466 / \text{rad}$$

$$C_{L_{\alpha,w}} = 5.0466 \cdot \frac{\pi}{180} = 0.0881 / \text{deg}$$

$$\rightarrow x_a = x_{cg} - x_{ac} = |25.64 - 27| = 1.36 \text{ ft aft of CG}$$

$$\bar{x}_a = \frac{1.36}{11.61} = 0.11714$$

$$M_w = C_{L_{\alpha,w}} \bar{x}_a = (5.0466 / \text{rad})(0.11714) = 0.5911 / \text{rad}$$

$$= (0.0881 / \text{deg})(0.11714) = 0.0103 / \text{deg}$$

• Horizontal Tail:

$$C_{L_\alpha} = \frac{2\pi AR}{2 + \sqrt{\frac{AR^2 \beta^2}{k^2} \left(1 + \frac{\tan^2 \Lambda_{c/4}}{\beta^2}\right) + 4}}$$

H.T. parameters $\rightarrow$

- $b = 36 \text{ ft}$
- $AR = 4.23$
- $S_{H.T.} = \frac{36^2}{4.23} = 305.38297$
- $\Lambda = 0.17$
- $\Lambda = 0^\circ$

$\hookrightarrow$ NACA 0012

$$C_{L_{\alpha,H.T.}} = \frac{2\pi(4.23)}{2 + \sqrt{\frac{4.23^2(0.985)^2}{1^2} (1 + 0) + 4}} = 4.01375 / \text{rad}$$

$$= 0.07005 / \text{deg}$$

$$\beta = \sqrt{1 - 0.7^2} = 0.985$$

$$k = \frac{a_0}{2\pi} \approx \frac{2\pi}{2\pi} = 1$$

• Downwash @ $Mach = 0 \rightarrow \left(\frac{\partial \varepsilon}{\partial \alpha} \right)_{h=0} = 0.283$

So $\left| \frac{\partial \varepsilon}{\partial \alpha} \right|_{h=0.17} = \left(\frac{\partial \varepsilon}{\partial \alpha} \right)_{h=0} \frac{(C_{L\alpha, w})_{h=0.17}}{(C_{L\alpha, w})_{h=0}}$

$$(C_{L\alpha, w})_{h=0} = \frac{2\pi AR}{2 + \sqrt{\frac{AR^2}{k^2} (1 + (\frac{2}{k})^2 C_{L\alpha})} + 4}$$

$$= \frac{2\pi (8.6)}{2 + \sqrt{\frac{8.6^2}{1} (1 + 0) + 4}} = 4.98964 \text{ /rad}$$

$$= 0.00867 \text{ /deg}$$

$$\left. \frac{\partial \varepsilon}{\partial \alpha} \right|_{h=0.17} = 0.283 \left(\frac{5.0464}{4.98964} \right) = 0.286595 \dots$$

$$\left(\frac{\partial \varepsilon}{\partial \alpha} \right)_{h=0.17} = 0.287$$

$$\eta = \frac{g_t}{g} \approx 1$$

since H.T is higher than wings

$$\bar{V} = \frac{S_{H.T}}{S_w} \frac{l_{H.T}}{\bar{c}_w} = \frac{306.383}{1023.95} \cdot \frac{31.36}{11.61} = 0.8082$$

$$l_{H.T} = x_{ac,T} - x_{cg} = 57 - 25.64 = 31.36 \text{ ft}$$

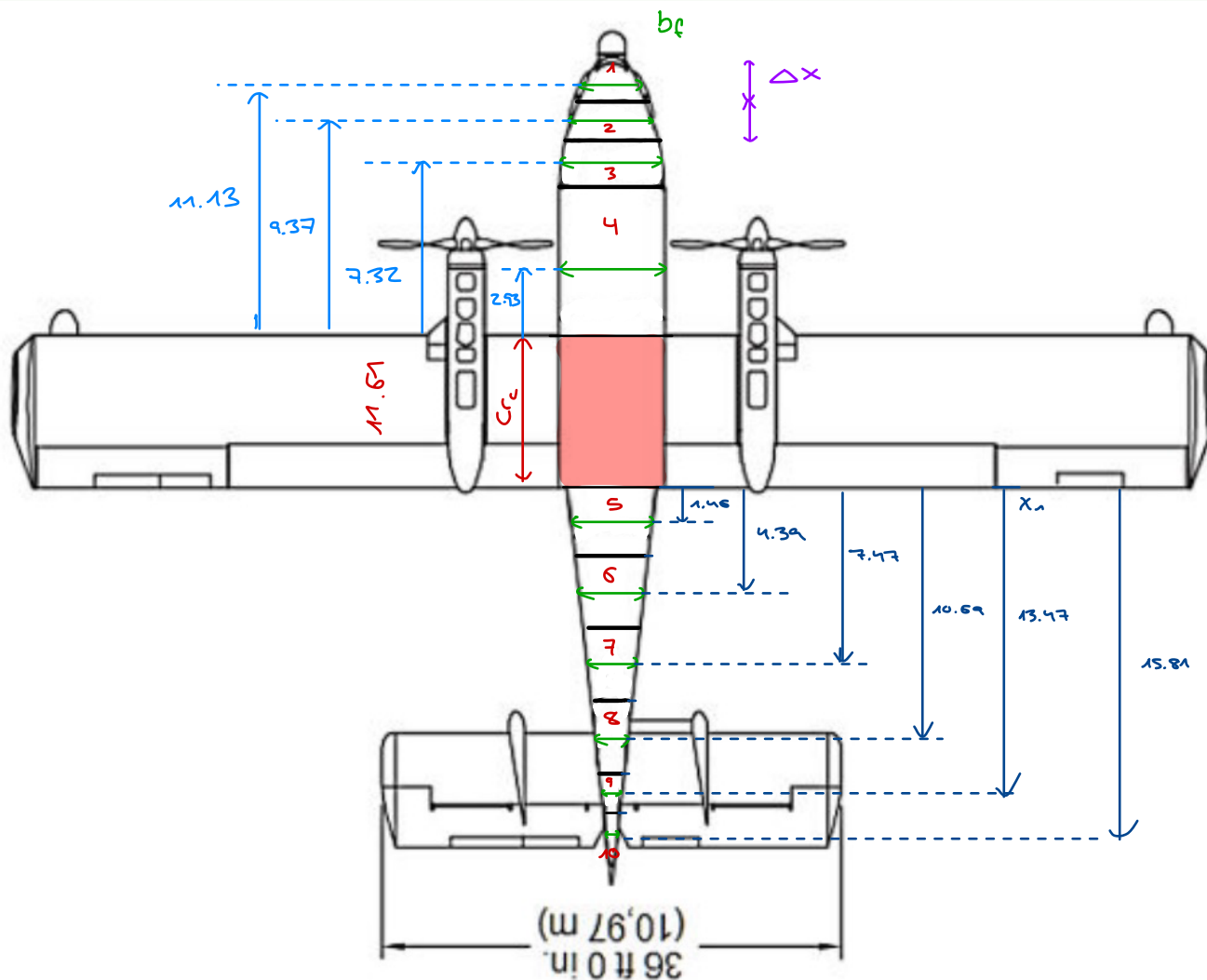
• Horizontal Tail:

$$(C_{m\alpha})_{H.T} = C_{L\alpha, H.T} \left(1 - \left. \frac{\partial \varepsilon}{\partial \alpha} \right|_{h=0.17} \right) \eta \bar{V}$$

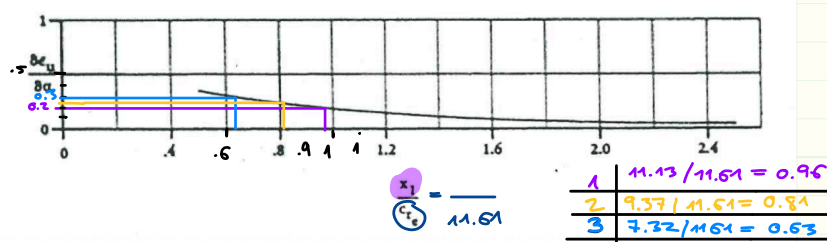
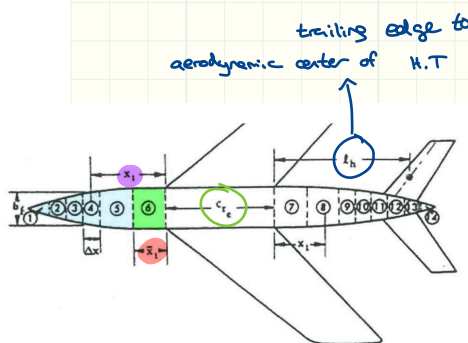
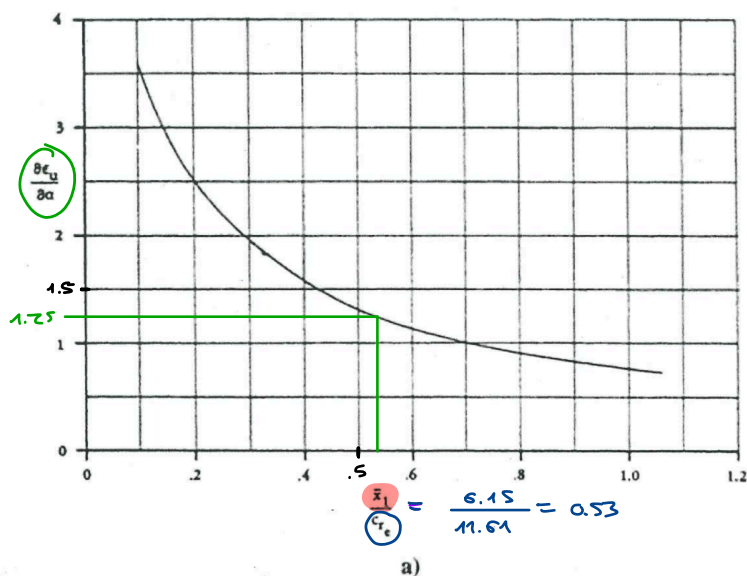
$$= 4.0127 (1 - 0.287) (1) (0.8082) = 2.31309 \dots \text{ /rad}$$

$$(C_{m\alpha})_{H.T} = 2.314 \text{ /rad} = 0.04038 \text{ /deg}$$

Multihop - Fuselage Contribution



Section	b_f^2	Δx	$\frac{\partial E_h}{\partial x}$	x_1	x_1/L_n	$(1 + \frac{\partial E_h}{\partial x})$
1	5.489	1.611	0.2			1.2
2	12.351	1.904	0.25			1.25
3	19.299	2.050	0.3			1.3
4	21.958	6.004	1.25			2.25
5	12.351	2.929		1.46	0.0688	0.019
6	7.741	2.929		4.39	0.2063	0.058
7	4.203	3.223		7.47	0.3508	0.099
8	2.144	2.929		10.62	0.5021	0.142
9	0.772	2.197		13.47	0.6328	0.179
10	0.086	2.636		15.81	0.7428	0.210



$$l_h = x_H - x_{TE} = 57 - [27 + 0.75(11.61)] = 57 - 35.71 = 21.29 \text{ ft}$$

$$\text{Downwash} = 1 + \frac{\partial \epsilon}{\partial \alpha} = \frac{x_f}{l_h} \left(1 - \frac{\partial \epsilon_h}{\partial \alpha} \right)$$

$$\frac{\partial \epsilon}{\partial \alpha} = 0.283 \quad (\text{from before}) \quad @ \quad M=0$$

$$1 - 0.283 = 0.717$$

Last column:

$$\frac{x_1}{l_h} (0.717) = 0.0688 (0.717)$$

$$(C_{m\alpha})_f = \frac{\pi}{2 S_w C_w} \sum_{i=1}^N b_f^2 \left(1 + \frac{\partial \epsilon_h}{\partial \alpha} \right) \Delta x$$

$$= \frac{\pi}{2 (1073.95) (11.61)} \left[(2.489)(1.6108)(1.2) + \dots \right]$$

$$(C_{m\alpha})_f = 0.051883 / \text{rad} = 0.000906 / \text{deg}$$

Total $C_{m\alpha}$

$$M_{\text{TOT}} = M_p + M_w + M_{\text{H.T}}$$

$$(C_{m\alpha})_{\text{TOT}} = C_{m\alpha,f} + C_{L\alpha,w} \bar{x}_a - C_{L\alpha,t} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \eta \bar{v}$$

Part	$C_{m\alpha}$ (rad)	$C_{m\alpha}$ (deg)
Fuselage	0.051883	0.00091
Wing	0.5911	0.01032
H.T	-2.314	0.04038
Total	-1.671	-0.02915

$$(C_{m\alpha})_{\text{TOT}} = -1.671 / \text{rad} = -0.0292 / \text{deg}$$